

COSC 594/650

Descent methods

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From optimality to algorithms

$$\begin{array}{ll} \underset{x}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

■ LOCAL OPTIMALITY

$$\nabla f(x) = 0$$

■ GLOBAL OPTIMALITY

CONVEXITY:

Stationary points $\equiv$ global minimizers

QUESTION: How to find those minimizers?

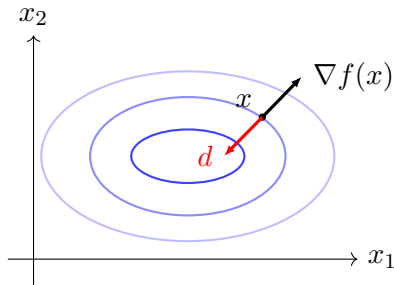
Descent direction

Non-stationary point: $\nabla f(x) \neq 0$

■ DESCENT DIRECTION

$$\nabla f(x)^\top d < 0$$

★ $f(x + \alpha d) < f(x)$, sufficient small $\alpha > 0$ – function descending



Iterative descent template

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Direction: d_k

Stepsize: $\alpha_k > 0$

An algorithm is a direction rule plus a step-size rule

■ DIRECTIONS

- ★ $d_k = -\nabla f(x_k)$ – steepest descent
- ★ $d_k = -D_k \nabla f(x_k), D_k \succ 0$ – diagonally scaled descent
- ★ $d_k = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$ – Newton direction

Iterative descent template

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Direction: d_k

Stepsize: $\alpha_k > 0$

An algorithm is a direction rule plus a step-size rule

■ STEP-SIZES

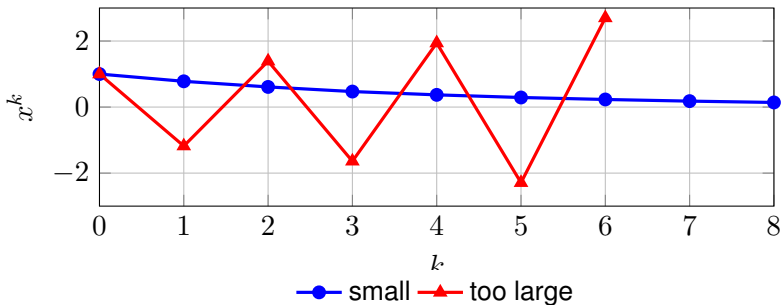
- ★ $\alpha_k = \alpha$ – constant
- ★ $\alpha_k \in \operatorname{argmin}_{\alpha \geq 0} f(x_k + \alpha d_k)$ – exact minimization
- ★ $\alpha_k \in \operatorname{argmin}_{\alpha \in (0, \bar{\alpha}]} f(x_k + \alpha d_k)$ – limited minimization
- ★ $\alpha_k \rightarrow 0, \sum_k \alpha_k = \infty$ – diminishing

Constant step-size: Stability matters

Function: $f(x) = \frac{1}{2}x^2$

Steepest descent: $d_k = -x_k$

Iterative update: $x_{k+1} = (1 - \alpha)x_k, k = 0, 1, 2, \dots$



Stability condition: $0 < \alpha < 2$

Exact line search on a quadratic

Quadratic function: $f(x) = \frac{1}{2}x^\top Qx + b^\top x, Q = Q^\top \succ 0$

Steepest descent: $d_k = -Qx_k - b$

Iterative update: $x_{k+1} = (I - \alpha_k Q)x_k - \alpha_k b, k = 0, 1, 2, \dots$

■ LINE SEARCH

$$\alpha_k \in \operatorname{argmin}_{\alpha \geq 0} f(x_k + \alpha d_k) = f(x_k) - \alpha (d_k)^\top d_k + \frac{\alpha^2}{2} d_k^\top Q d_k$$

$$\implies \alpha_k = \frac{(d_k)^\top d_k}{(d_k)^\top Q d_k}$$

Armijo backtracking

Actual decrease: $f(x_k) - f(x_k + \alpha d_k)$

Predicted decrease: $-\nabla f(x_k)^\top d_k$

■ SUFFICIENT DECREASE

$$f(x_k) - f(x_k + \alpha d_k) \geq -\sigma \nabla f(x_k)^\top d_k, \quad \sigma \in (0, 1)$$

■ BACKTRACK

$$\alpha_k = \max \left\{ \bar{\alpha} \beta^i \mid f(x_k) - f(x_k + \bar{\alpha} \beta^i d_k) \geq -\sigma \bar{\alpha} \beta^i \nabla f(x_k)^\top d_k, i = 0, 1, 2, \dots \right\}$$

★ $\bar{\alpha}$ – initial step-size

★ $\beta \in (0, 1)$ – diminishing rate

Well-defined step-size if $\nabla f(x_k)^\top d_k < 0$

What does convergence analysis mean?

■ STATIONARITY

$$\nabla f(x^\star) = 0$$

■ CONVERGENCE METRIC

$$e(x_k) = \|x_k - x^\star\| \quad \text{or} \quad e(x_k) = f(x_k) - f(x^\star)$$

■ ITERATION COMPLEXITY

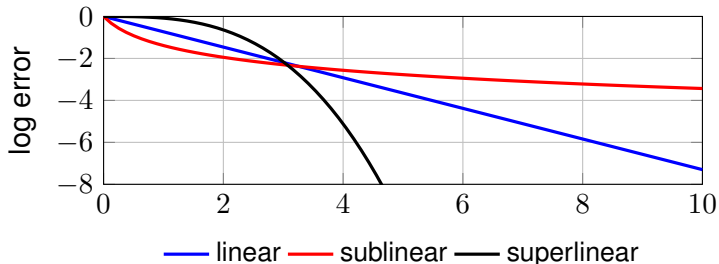
$$\# \min \{k \mid e(x_k) \leq \epsilon, k = 0, 1, 2, \dots\}$$

These can be separate questions
One may answer one without answering all three!

Asymptotic rate

$$\rho = \limsup_{k \rightarrow \infty} \frac{e(x_{k+1})}{e(x_k)}, \quad x_k \rightarrow x^\star$$

- ★ $0 < \rho < 1$ – linear rate
- ★ $\rho = 1$ – typically sublinear rate
- ★ $\rho = 0$ – superlinear convergence



Gradient-related condition

Subsequence: $\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x}$, a non-stationary point

Directions: $\{d_k\}_{k \in \mathcal{K}}$

■ GRADIENT-RELATED DIRECTIONS

- ★ $\{d_k\}_{k \in \mathcal{K}}$ – bounded directions
- ★ $\limsup_{k \rightarrow \infty, k \in \mathcal{K}} \nabla f(x_k)^\top d_k < 0$ – subsequence limit

■ SCALED GRADIENT DIRECTIONS

$$d_k = -D_k \nabla f(x_k), \quad \underline{\gamma} I \preceq D_k \preceq \bar{\gamma} I$$

$$\implies \nabla f(x_k)^\top d_k \leq -\underline{\gamma} \|\nabla f(x_k)\|^2$$

Global convergence to stationary points

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rules: constant, diminishing, Armijo backtracking

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow \bar{x} \implies \bar{x} \text{ is a stationary point}$

Every limit point $x^\star$ is stationary: $\nabla f(x^\star) = 0$

Convexity is not required!

Why the stationarity theorem works?

■ DESCENT LEMMA

$$f(x+h) - f(x) \leq h^\top \nabla f(x) + \frac{L}{2} \|h\|^2, \quad \forall h \in \mathbb{R}^n$$

★ $f(x_k + \alpha_k d_k) - f(x_k) \leq \alpha_k \nabla f(x_k)^\top d_k + \frac{(\alpha_k)^2 L}{2} \|d_k\|^2$ – one-step improvement

PROOF: Lipschitz continuity

Stationarity theorem: Diminishing step-size

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: $\alpha_k \rightarrow 0, \sum_k \alpha_k = \infty$

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow x^* \implies x^* \text{ is a stationary point}$

PROOF: By contradiction

Stationarity theorem: Armijo backtracking

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha_k d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: Armijo backtracking

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow x^* \implies x^*$ is a stationary point

PROOF: By contradiction

Stationarity theorem: Constant step-size

Lipschitz continuous gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^n$

Iterative update: $x_{k+1} = x_k + \alpha d_k, k = 0, 1, 2, \dots$

Gradient-related directions: $\{d_k\}_{k \in \mathcal{K}}$

Step-size rule: $0 < \epsilon \leq \alpha \leq \frac{(2 - \epsilon)|\nabla f(x_k)^\top d_k|}{L\|d_k\|^2}$

■ STATIONARY-POINT CONVERGENCE

$\{x_k\}_{k \in \mathcal{K}} \rightarrow x^\star \implies x^\star \text{ is a stationary point}$

PROOF: By contradiction

Questions?

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